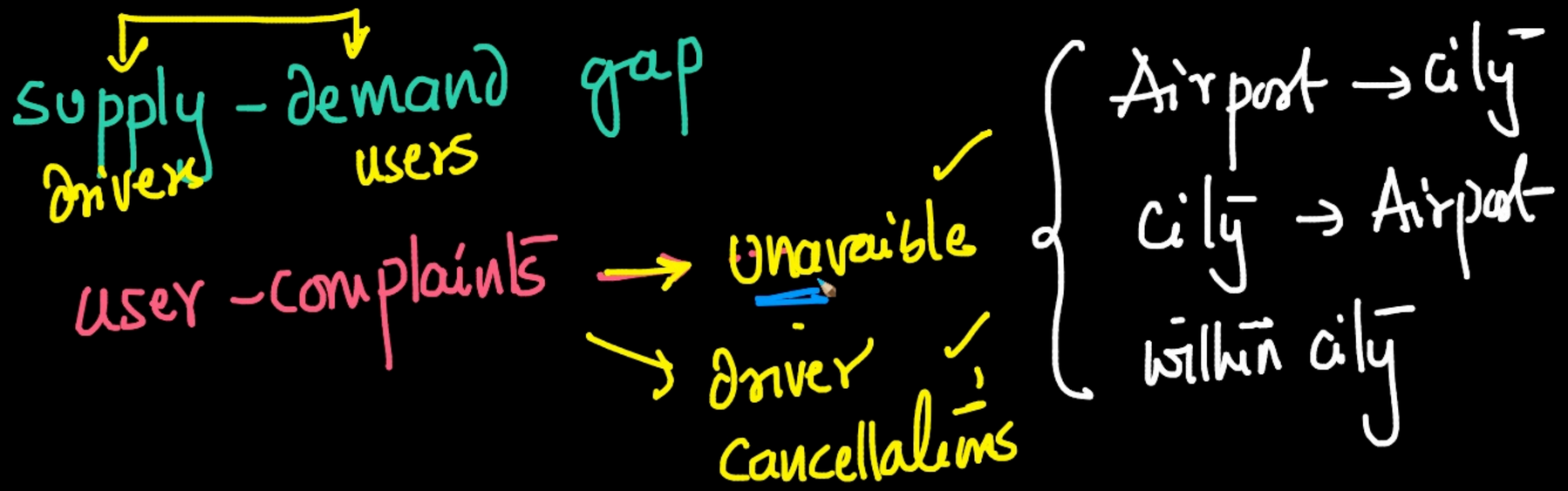
Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:04

RCA - 2 → Uber + code → Excel, Tableau, Python

class starts @ 9:03 PM

Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:12

Problem:



Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:10

Hypothesis / ques:

1. Is this problem limited to a few hotspots?
   - ↳ Region
2. Driver economics — problem → Internal (other)
   - ↳ Revenue metric
3. Time-windows limited → Time
   - ↳ Hour of day
   - → Holiday; Friday

Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:12

(4) Is cancellation due to a specific type of destination → Region

(5) Weather — External

⋮

CRIED ✓

↓

TROPICS

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Search results - Google Drive | RCA-2.ipynb - Colaboratory | uber-data.csv - Google Drive | pandas.to_datetime — pandas

colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=YO5HjFbZSWFV

RCA-2.ipynb

File Edit View Insert Runtime Tools Help All changes saved

Comment Share

+ Code + Text

RAM Disk

[6] 27s

Choose files uber-data.csv

- **uber-data.csv**(text/csv) - 395061 bytes, last modified: 03/05/2023 - 100% done

```
Saving uber-data.csv to uber-data (1).csv
```

1s

```
# Loading the dataset -

df = pd.read_csv('uber-data.csv', parse_dates=[4,5], dayfirst=True, na_values="NA")
df.head()
```

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	619	Airport	1.0	Trip Completed	2016-07-11 11:51:00	2016-07-11 13:00:00
1	867	Airport	1.0	Trip Completed	2016-07-11 17:57:00	2016-07-11 18:47:00
2	1807	City	1.0	Trip Completed	2016-07-12 09:17:00	2016-07-12 09:58:00
3	2532	Airport	1.0	Trip Completed	2016-07-12 21:08:00	2016-07-12 22:03:00
4	3112	City	1.0	Trip Completed	2016-07-13 08:33:16	2016-07-13 09:25:47

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colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=YO5HjFbZSWFV

RCA-2.ipynb

File Edit View Insert Runtime Tools Help All changes saved

Comment Share

+ Code + Text

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[6] 27s

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6 / 6

Search results - Google Drive | RCA-2.ipynb - Colaboratory | uber-data.csv - Google Drive | pandas.to_datetime — pandas | pandas.read_csv — pandas 2.

colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=aitYOXYXUQ2q

+ Code + Text

RAM Disk

RCA

↳ { cancellations
not Available

[5] 0s
```
# Importing required libraries -

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[6] 27s
```
from google.colab import files
uploaded = files.upload()
```

Choose files uber-data.csv

- **uber-data.csv**(text/csv) - 395061 bytes, last modified: 03/05/2023 - 100% done

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[7] 1s
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3	2532	Airport	1.0	Trip Completed	2016-07-12 21:08:00	2016-07-12 22:03:00
4	3112	City				

Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:24

[11] 0s

```
# Checking for duplicate rows -

print("No. of duplicate rows: ", df.duplicated().sum())
```

```
No. of duplicate rows:  0
```

```
df["RequestHour"] = df["Request timestamp"].dt.hour
```

Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:27

Search results - Google Drive | RCA-2.ipynb - Colaboratory | uber-data.csv - Google Drive | pandas.to_datetime — pandas | pandas.read_csv — pandas 2.

colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=Bd64ZDicVB3M

Update

+ Code + Text

RAM Disk

```
[12] df["RequestHour"] = df["Request timestamp"].dt.hour
```

0s

```
[13] df.head()
```

0s

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp	RequestHour
0	619	Airport	1.0	Trip Completed	2016-07-11 11:51:00	2016-07-11 13:00:00	11
1	867	Airport	1.0	Trip Completed	2016-07-11 17:57:00	2016-07-11 18:47:00	17
2	1807	City	1.0	Trip Completed	2016-07-12 09:17:00	2016-07-12 09:58:00	9
3	2532	Airport	1.0	Trip Completed	2016-07-12 21:08:00	2016-07-12 22:03:00	21
4	3112	City	1.0	Trip Completed	2016-07-13 08:33:16	2016-07-13 09:25:47	8

0-4 4-9 9-16 16-21 21+

```
# Separate 5 different timeslots from the Hour - Dawn, Early Morning, Noon, Late Evening, Night -

df["TimeSlot"] = df["RequestHour"].apply(lambda x: "Dawn" if x<=4 else ("Early Morning"
                                                    if x<=9 else ("Noon"
                                                        if x<=16 else ("Late Evening"
                                                            if x<=21 else "Night"))))
```

9 / 9

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colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=Bd64ZDicVB3M
Update
+ Code + Text
RAM
Disk

[12] `df["RequestHour"] = df["Request timestamp"].dt.hour`

0s

[13] `df.head()`

0s

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp	RequestHour
0	619	Airport	1.0	Trip Completed	2016-07-11 11:51:00	2016-07-11 13:00:00	11
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                                                      if x<=16 else ("Late Evening"
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```

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Search results - Google Drive | RCA-2.ipynb - Colaboratory | uber-data.csv - Google Drive | pandas.to_datetime — pandas | pandas.read_csv — pandas 2.

colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=Bd64ZDicVB3M

Update

+ Code + Text

RAM Disk

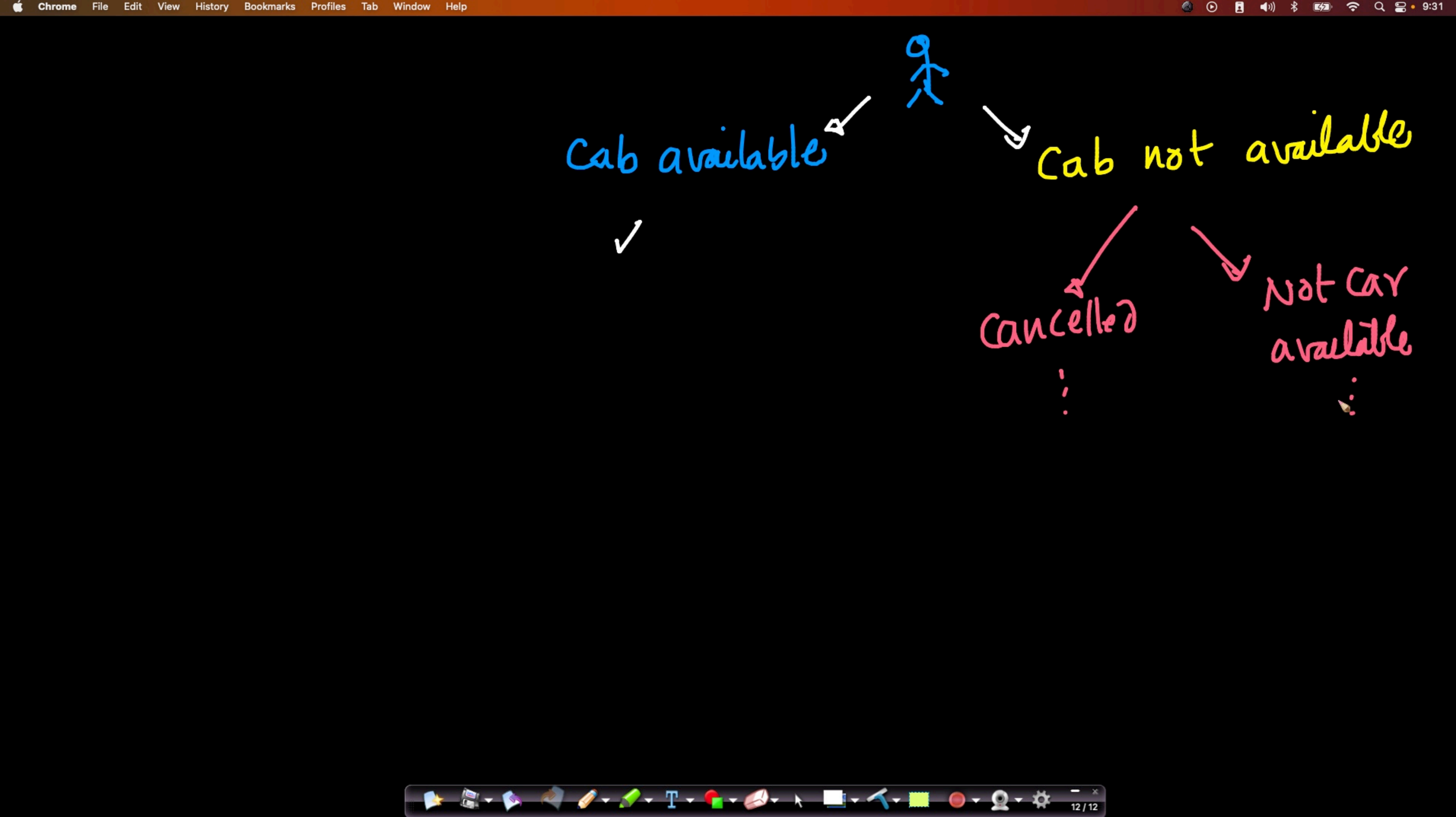
[12] 0s
```
df["RequestHour"] = df["Request timestamp"].dt.hour
```

[13] 0s
```
df.head()
```

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp	RequestHour
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```
# Separate 5 different timeslots from the Hour - Dawn, Early Morning, Noon, Late Evening, Night -

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                                                 if x<=9 else ("Noon"
                                                      if x<=16 else ("Late Evening"
                                                           if x<=21 else "Night"))))
```


Cab available
Cab not available
Cancelled
Not cav available

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Search results - Google Drive | RCA-2.ipynb - Colaboratory | uber-data.csv - Google Drive | pandas.to_datetime — pandas | pandas.read_csv — pandas 2.
colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=iaacfQLaWAhK
+ Code + Text
RAM
Disk

[15] 0s

1710	5512	Airport	180.0	Trip Completed	2016-07-15 05:10:25	2016-07-15 05:42:24	5	Early Morning
2694	3867	City	284.0	Trip Completed	2016-07-13 21:45:14	2016-07-13 22:24:40	21	Late Evening
6489	6285	Airport	NaN	No Cars Available	2016-07-15 18:18:28	NaT	18	Late Evening
6023	5135	Airport	NaN	No Cars Available	2016-07-14 20:16:25	NaT	20	Late Evening
508	1958	Airport	52.0	Trip Completed	2016-07-12 11:41:00	2016-07-12 12:35:00	11	Noon
2813	4121	Airport	298.0	Trip Completed	2016-07-14 04:29:45	2016-07-14 05:07:30	4	Dawn
3972	1511	City	272.0	Cancelled	2016-07-12 05:40:00	NaT	5	Early Morning

[16] 0s

```
# Distinguish the Supply-Demand Gap by a new variable Cab Availability where Supply is when Trip is Completed, all else is Demand -

df["Cab Availability"] = df["Status"].apply(lambda x: "Available" if x=="Trip Completed" else "Not Available")
```

0s

```
df['Cab Availability'].value_counts(normalize=True)*100
```

```
Not Available    58.028169
Available        41.971831
Name: Cab Availability, dtype: float64
```

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Search results - Google Drive | RCA-2.ipynb - Colaboratory | uber-data.csv - Google Drive | pandas.to_datetime — pandas | pandas.read_csv — pandas 2.
colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=WYV8WHVdBJRe
Update
+ Code + Text
RAM
Disk

```
 2   Driver id          float64
 3   Status             object
 4   Request timestamp  datetime64[ns]
 5   Drop timestamp     datetime64[ns]
dtypes: datetime64[ns](2), float64(1), int64(1), object(2)
memory usage: 316.3+ KB
```

```
# Checking for null values -

df.isnull().sum() / len(df) * 100
```

```
Request id           0.000000
Pickup point         0.000000
Driver id           39.388362
Status               0.000000
Request timestamp    0.000000
Drop timestamp      58.028169
dtype: float64
```

why is this an issue for this analysis

```
# Checking for duplicate rows -

print("No. of duplicate rows: ", df.duplicated().sum())
```

```
No. of duplicate rows:  0
```

```
df["RequestHour"] = df["Request timestamp"].dt.hour
```

```
df.head()
```

Request id Pickup point D

14 / 14

```
[16] df["Cab Availability"] = df["Status"].apply(lambda x: "Available" if x=="Trip Completed" else "Not Available")
```

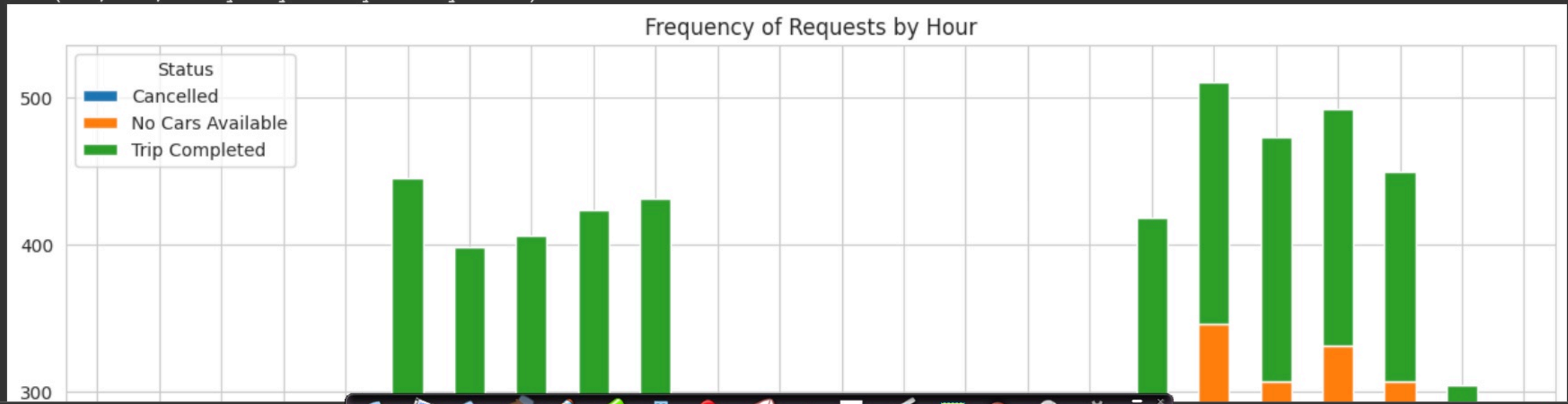
```
[17] df['Cab Availability'].value_counts(normalize=True)*100
```

```
Not Available    58.028169
Available        41.971831
Name: Cab Availability, dtype: float64
```

```
# Frequency of Requests by Hour -

df.groupby(['RequestHour','Status']).size().unstack().plot(kind='bar', stacked=True, figsize=(15, 8))
plt.title('Frequency of Requests by Hour')
```

```
Text(0.5, 1.0, 'Frequency of Requests by Hour')
```



pandas.DataFrame.astype
pandas.DataFrame.convert_dtypes
pandas.DataFrame.infer_objects
pandas.DataFrame.copy
pandas.DataFrame.bool
pandas.DataFrame.head
pandas.DataFrame.at
pandas.DataFrame.iat
pandas.DataFrame.loc
pandas.DataFrame.iloc
pandas.DataFrame.insert
pandas.DataFrame.__iter__
pandas.DataFrame.items
pandas.DataFrame.keys
pandas.DataFrame.iterrows
pandas.DataFrame.itertuples
pandas.DataFrame.pop
pandas.DataFrame.tail
pandas.DataFrame.xs
pandas.DataFrame.get

Reference the user guide for more examples.

Show Source

Examples

```
>>> index = pd.MultiIndex.from_tuples([('one', 'a'), ('one', 'b'),
...                                    ('two', 'a'), ('two', 'b')])
>>> s = pd.Series(np.arange(1.0, 5.0), index=index)
>>> s
one  a   1.0
     b   2.0
two  a   3.0
     b   4.0
dtype: float64
```

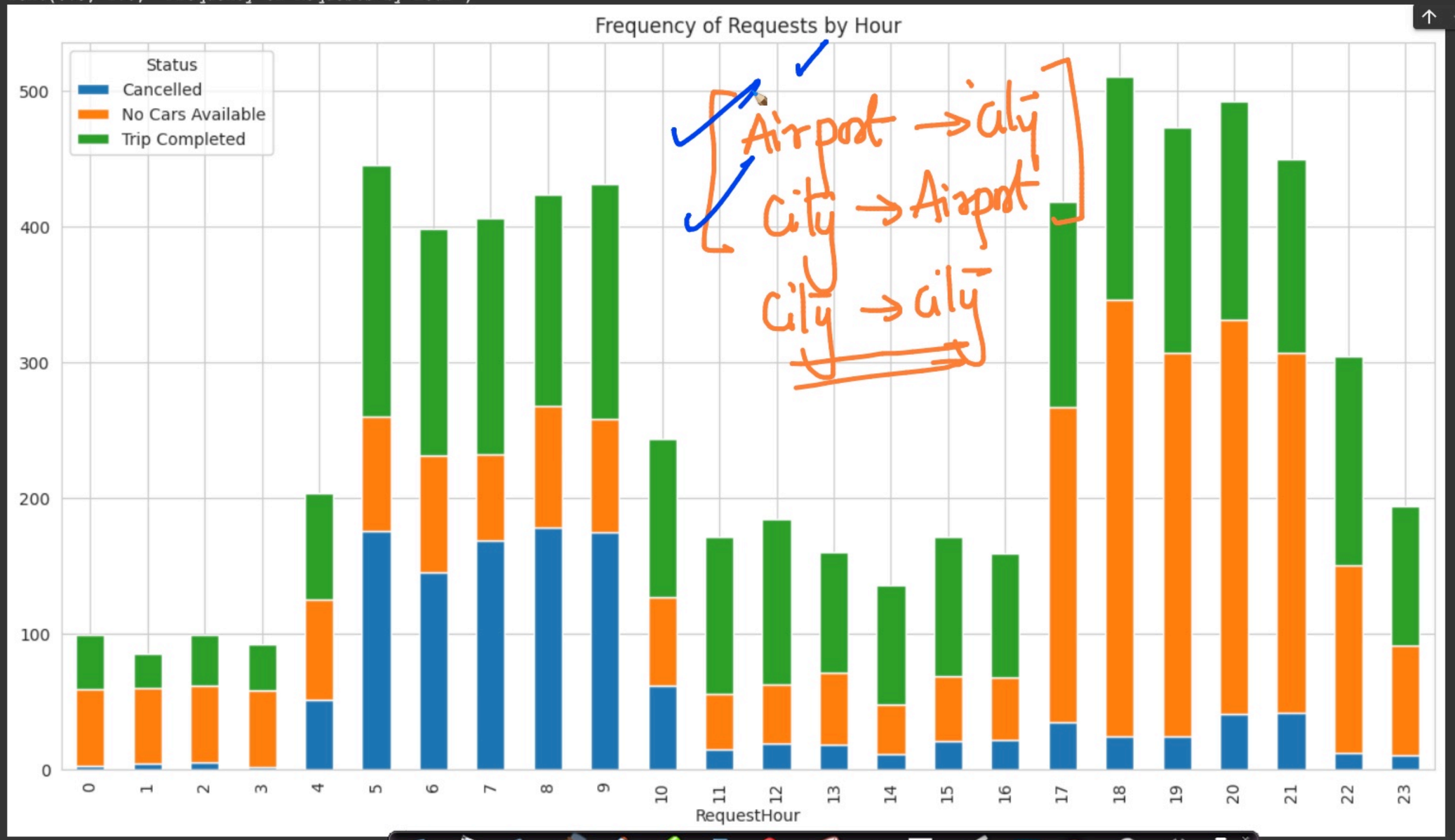
```
>>> s.unstack(level=-1)
     a   b
one  1.0  2.0
two  3.0  4.0
```

crosstab / pivottable / groupby

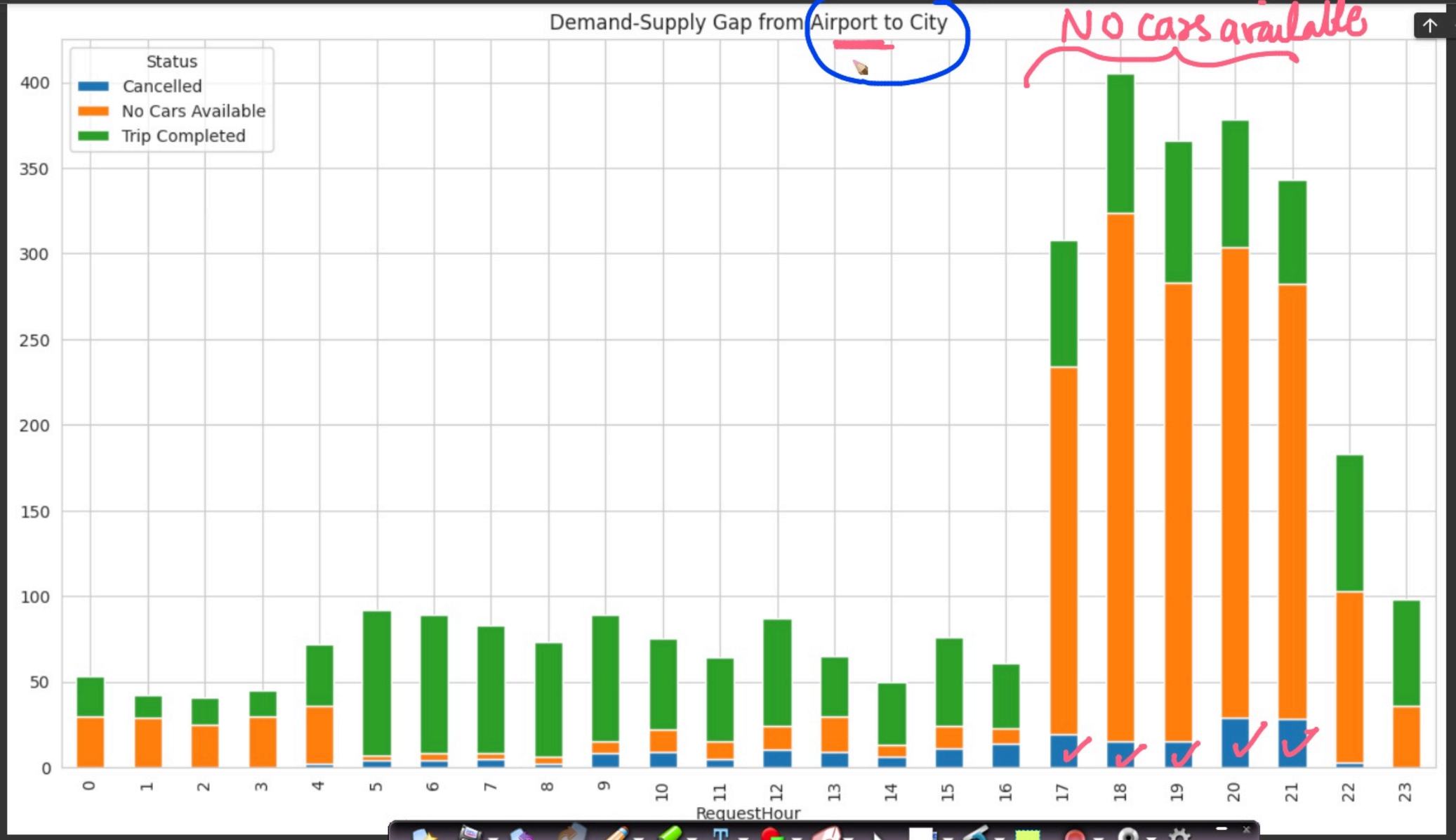
```
>>> s.unstack(level=0)
   one  two
a  1.0   3.0
b  2.0   4.0
```

```
>>> df = s.unstack(level=0)
>>> df.unstack()
```


Frequency of Requests by Hour
Status
Cancelled
No Cars Available
Trip Completed
RequestHour
lots of cancellations
cars not available
1
(WHY?)
2
solution


Frequency of Requests by Hour
Status
Cancelled
No Cars Available
Trip Completed
500
400
300
200
100
0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
RequestHour
Airport → city
city → Airport
city → city

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colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=ubaLbhXrZaeA
Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.
+ Code + Text
RAM
Disk



19 / 19

Search results - Google Drive | RCA-2.ipynb - Colaboratory | uber-data.csv - Google Drive | pandas.DataFrame.unstack — | pandas.read_csv — pandas 2.

colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=ubaLbhXrZaeA

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.

+ Code + Text

RAM Disk

Let's assume:

Airport → city

city → airport

all trips in this dataset

[5] 0s
```
# Importing required libraries -

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[6] 27s
```
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Choose files uber-data.csv

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4	3112	City	1.0	Trip Completed	2016-07-13 08:33:16	2016-07-13 09:25:47

0s
```
# Shape of the dataset -

print("No. of rows: {}".format(df.shape[0]))
print("No. of cols: {}".format(df.shap
```

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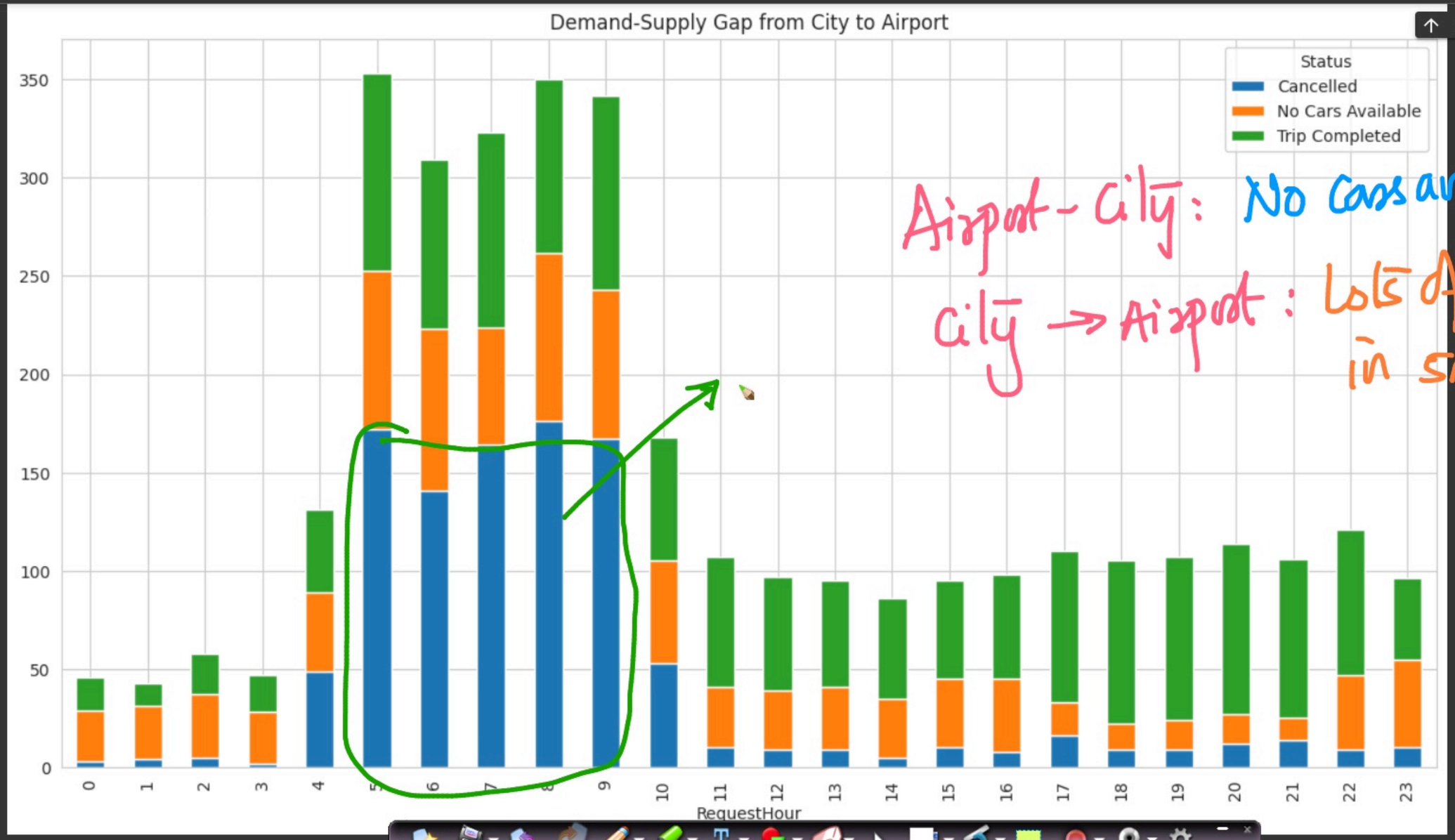
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colab.research.google.com/drive/1YGAz1airu9igRJnqmghVEPCzgvE_8yiz#scrollTo=dQMXYZoda8-j

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.

+ Code + Text

RAM Disk



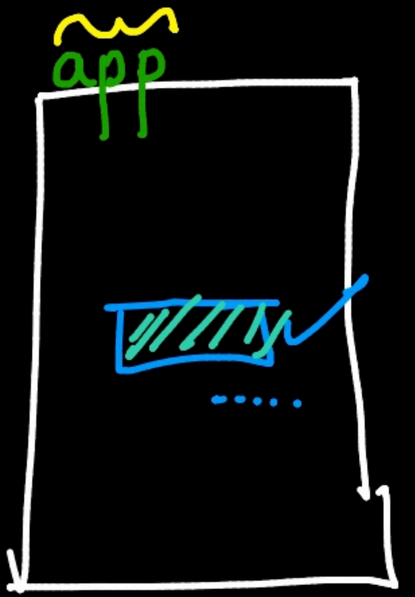
21 / 21

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:24

Banking / fintech ✓

Chatbot ✓

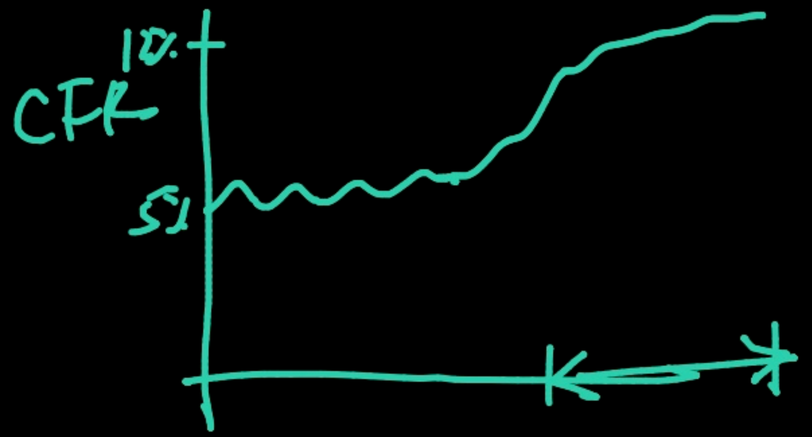
Chatbot failure rate has increased from 5% to 12% in last 5 days

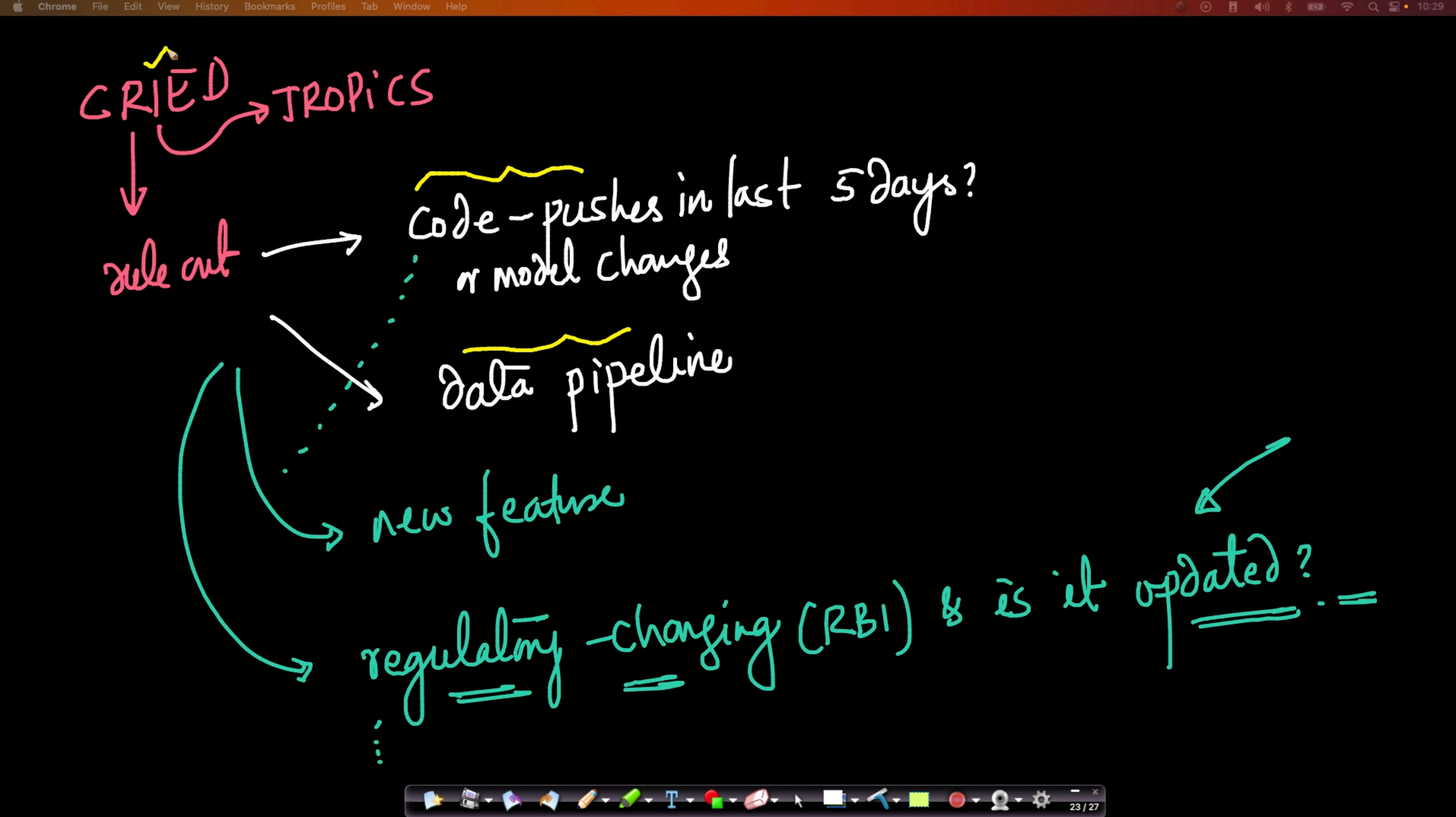


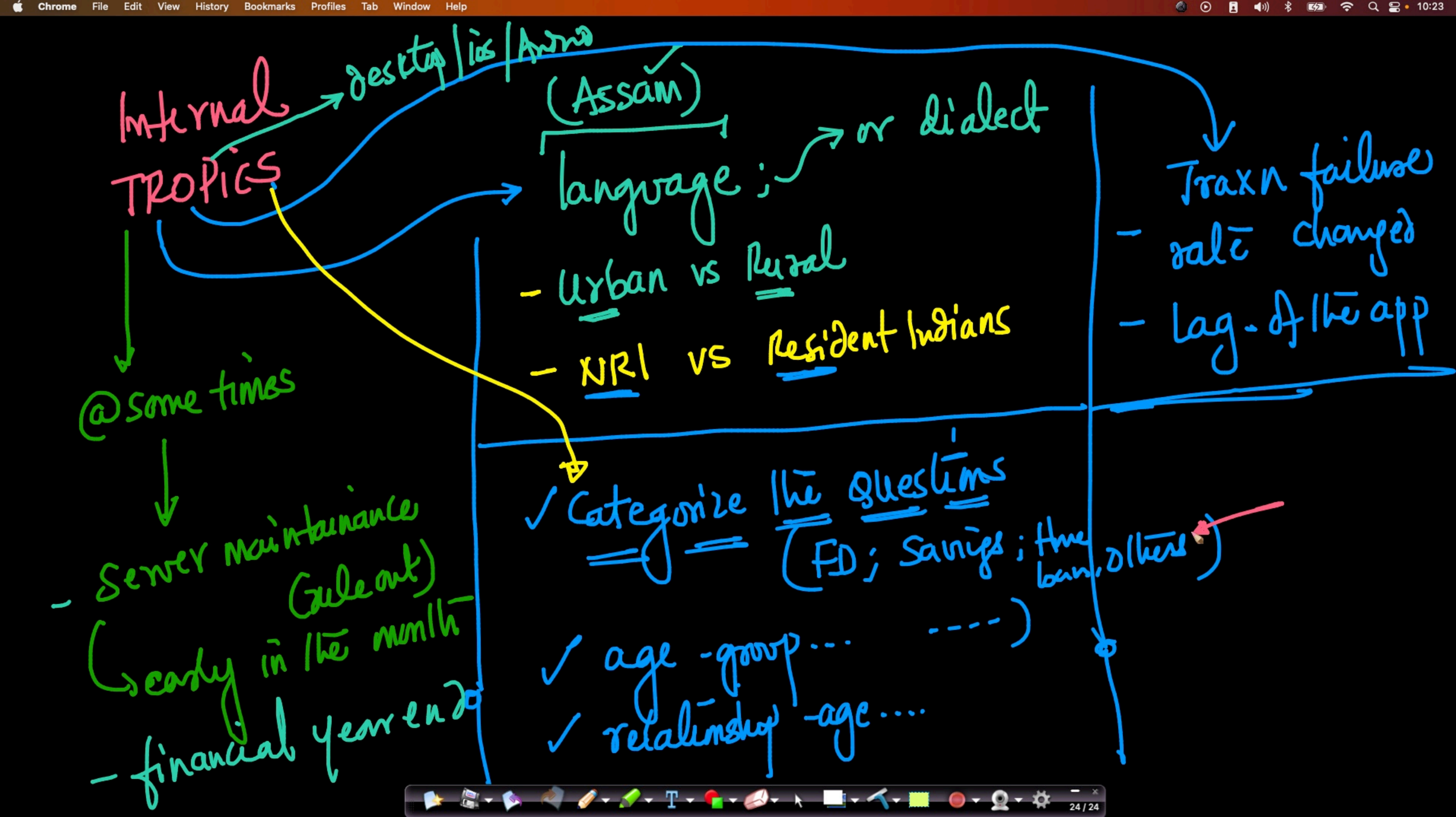
Chatbot – NLP

HDFC - Eva ✓

① Clarify: Metric = CFR = # users who asked to talk to a CSA / Total # users of the chatbot

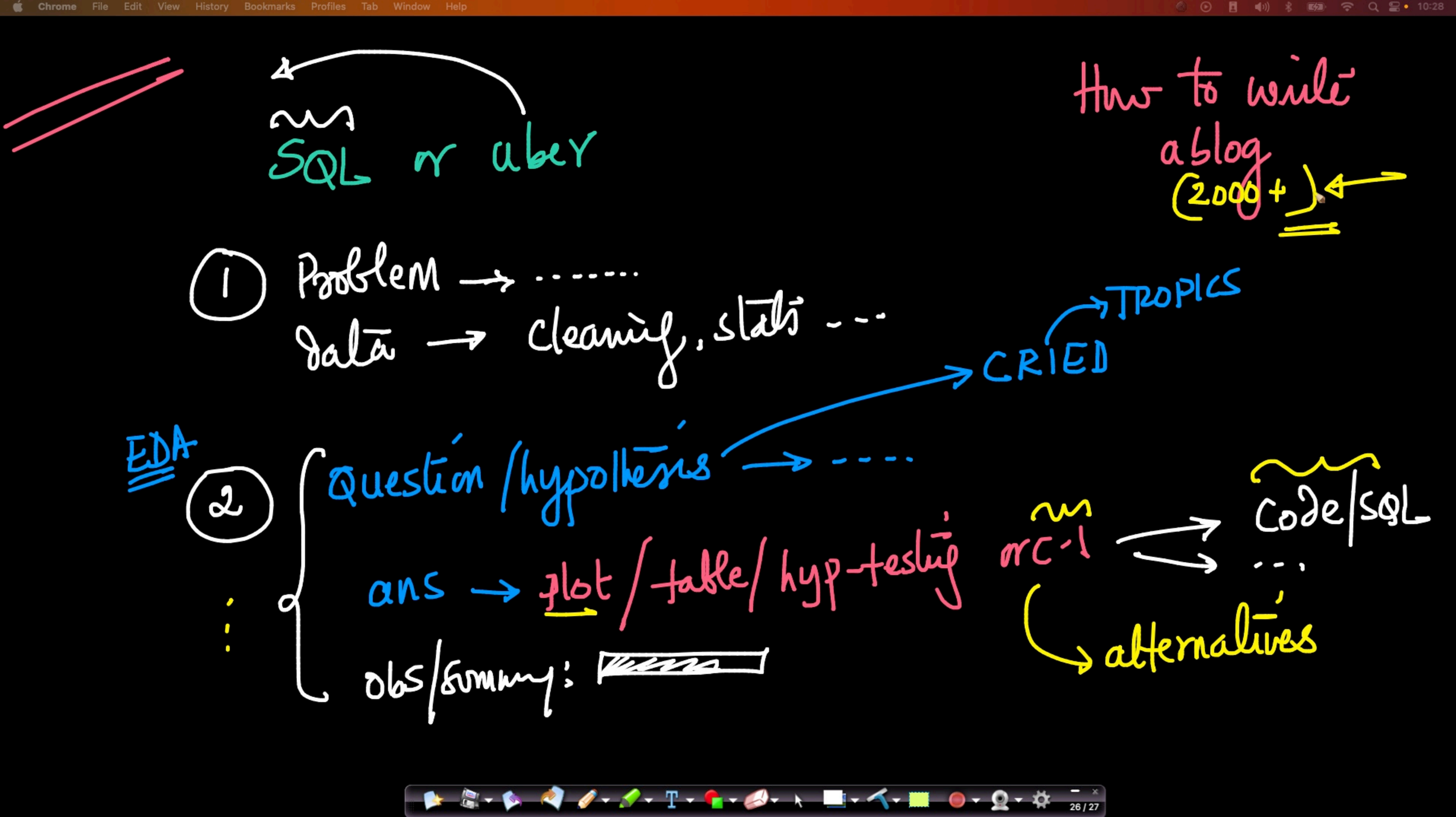


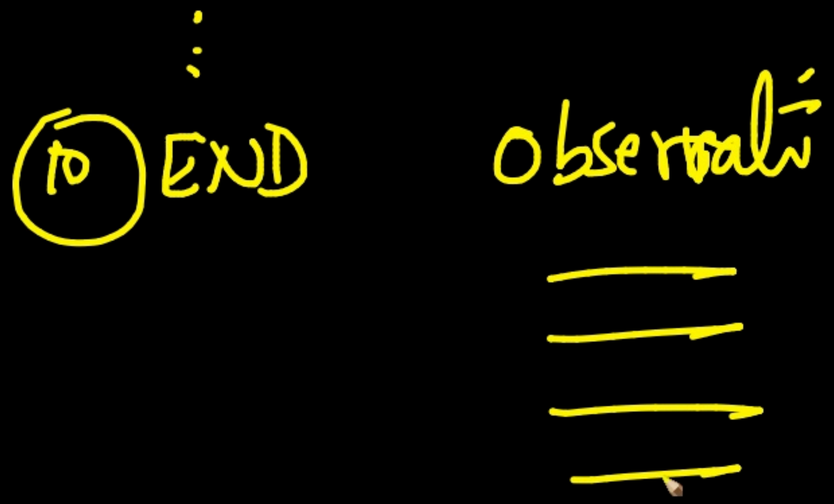

CRIED → TROPICS
rule out
code - pushes in last 5 days?
or model changes
data pipeline
new feature
regularly - changing (RBI) & is it updated?


Internal
TROPICS
desktop | ios | Android
(Assam)
language ; → or dialect
- Urban vs Rural
- NRI vs Resident Indians
Traxn failure
- sale changes
- lag - of the app
@ sometimes
- Server maintainance
(Late out)
(early in the month
- financial year end
✓ Categorize the questions
(FD ; Savings ; Home loan, others
✓ age - group ...
✓ relationship - age ....

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:24




SQL or ubeY
How to write a blog (2000+)
1
Problem → .....
Data → cleaning, stats ...
CRIED
TROPICS
EDA
2
Question / hypothesis → ....
ans → plot / table / hyp-testing
obs / summary:
src-1
code / SQL
...
alternatives


10
END
Observable

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RCA-2.ipynb

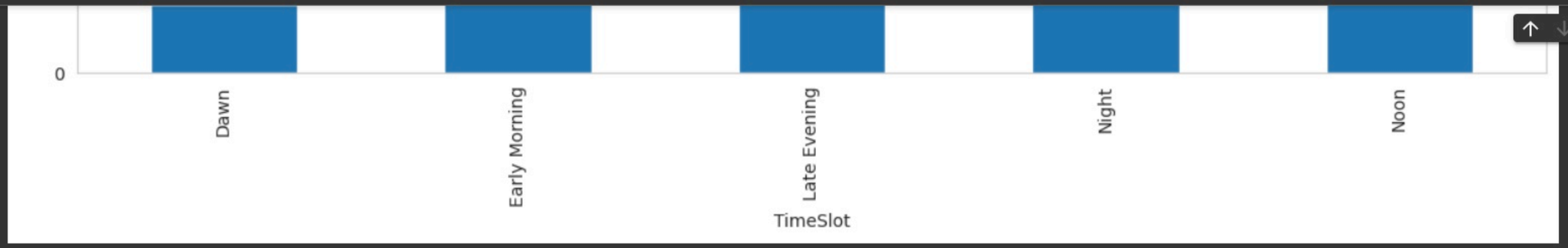
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